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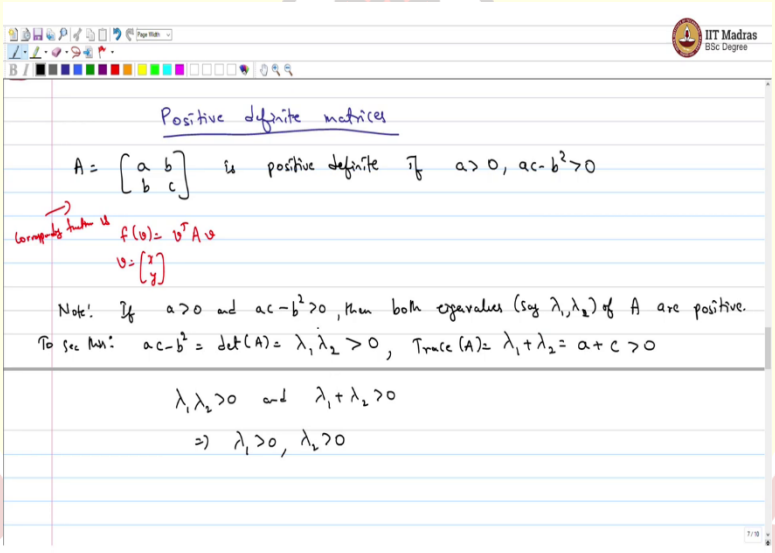
ONLINE DEGREE

Machine Learning Foundation
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Positive Definite Matrices

Welcome to lecture 2, in week 6 of Machine Learning Foundations. In this lecture, we will continue the discussion of positive definiteness. We started this in the context of functions in the last lecture. In this lecture, we will define the notion of positive definiteness in the context of a real symmetric matrix.

We will define what it means to say, real symmetric matrix is positive definite. And then we will also see what this definition is equivalent to. And we will see the equivalent, we will see two equal and ways to define a positive definite matrix, we will show this equivalence rigorously.

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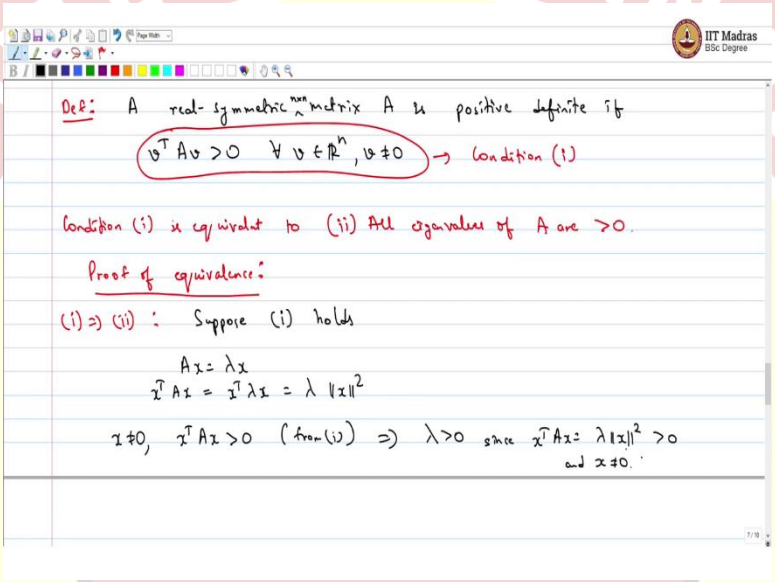
Now, we move on to the main topic, which is Positive Definite Matrices. We have seen what a positive definite function is. And I mean take a matrix like this. We saw this in the dot product. Now this is positive definite. If $a > 0$ an $ac - b^2 > 0$. Now, this is positive definite because I mean, the corresponding function here the corresponding function is f(v), where v is $\begin{bmatrix} x \\ y \end{bmatrix}$.

So, the function f is positive definite. If this is equivalent to saying some of this matrix is positive definite. And this function being positive definite is just connected to $a > 0$ and $ac - b^2 > 0$. And the important note is if $a > 0$ and $ac - b^2 > 0$, then both eigenvalues, say λ_1, λ_2 of A are positive.

I mean, to see this, ac - b^2 is equal to determinant of A is the same as $\lambda_1$into λ_2, and we know that this is greater than 0. And what is the trace of A = λ_1+ λ_2. The trace of a matrix is the sum of the diagonal entries, and also it is a sum of the eigenvalues. And so, we know that a + c is also, trace of this is this.

And this is clearly greater than 0, because we know that ac > b^2, so A > 0, a×c is definitely positive and since a is positive, so c is also positive. And that just means so it is very clear. So, what do you have is $\lambda_1 \times \lambda_2 > 0$ and also $\lambda_1 + \lambda 2 > 0$ implies $\lambda_1 > 0$ and $\lambda 2$ is > 0.

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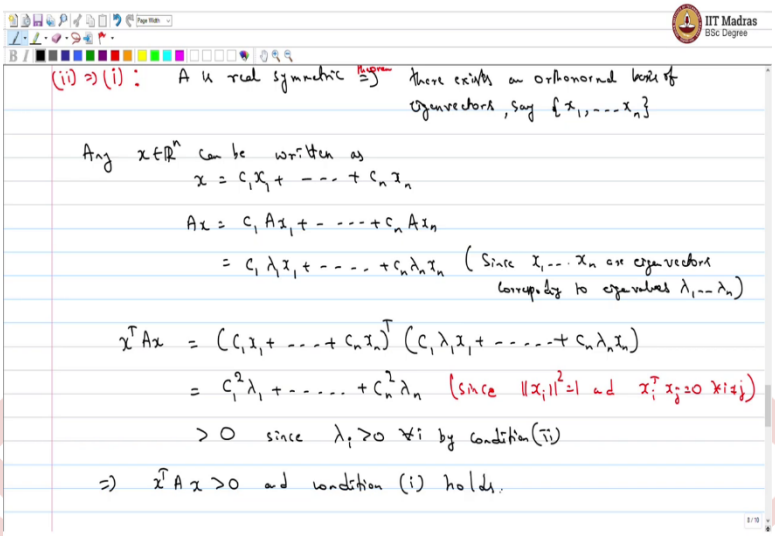


It is easy to infer this. Now, this is a property that gets generalized also like a real symmetric n×n matrix A is positive definite. If $v^T Av$ is > 0, for all v equivalent to $\mathbb{R}^n$, v is not the trivial vector. Now, let me call this say condition 1.

Now, condition 1 is equivalent to all eigenvalues of A are strictly positive. So, we could define, in particular, we could define saying real symmetric. Matrix is positive definite if all of its eigenvalues are greater than 0. And that is the same as saying condition 1. Proof of equivalence, first part, we will show 1 implies 2. Suppose 1 holds and take the eigen equation Ax = λx. If I do $x^T Ax$, $x^T \lambda x$ and just $\lambda \|x\|^2$.

Now, for x that is not equal to 0 because it is an eigenvector. And $x^T Ax > 0$ from 1. So, this implies λ > 0. Since $x^T Ax = \lambda \|x\|^2 > 0$, and x not equal to 0. So, since x is not equal to 0 and $x^T Ax$ is the same as $\lambda \|x\|^2$, so λ has to be > 0. Since x is not equal 0, so you have a positive quantity here and so λ can neither be 0 nor less than 0.

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Now, the reverse proof 2 implies 1. A is real symmetric, implies, this is from the spectral theorem. Implies there exists an orthonormal basis eigenvector, say, $\{x_{1,\ldots\ldots}$ until xn$\}$. Now, for any x, so any x belonging to $\mathbb{R}^n$ can be written using this, x1 to xn is a basis, so, $x = c_1x_1 + c_2x_2 \ldots\ldots\ldots$so on until c_n x_n. because x1 to xn is a basis. Now, $Ax = c_1Ax_1 + \ldots\ldots + c_nAx_n$, but these x_1 to x_n are eigenvectors, so $c_1\lambda_1x_1 + \ldots\ldots + c_n\lambda_nx_n$. Since x_1 to xn are eigenvectors, corresponding to eigenvalues λ_1 until λn.

Now, what is x^TAx ? x^TAx is $(c_1x_1 + \ldots\ldots$ +so on until $c_nx_n)^T$ into Ax, Ax is nothing but($c_1\lambda_1x_1 + \ldots\ldots\ldots + c_n\lambda_nx_n$). And this is equal to ($c_1^2\lambda_1 + \ldots\ldots$+ so on until $c_n^2\lambda$n) since norm $||Xi||^2 = 1$. And x_i^Txj = 0 (for i not equal to j).

Now, this is clearly greater than 0 since $\lambda_i > 0$ for all i by assumption 2, so. We started with 2 and we are trying to infer 1, so we are assuming that all the λ is greater than 0 and $c_1^2, \ldots c_n^2$ all greater than 0. And so, in place $x^TAx > 0$ and condition 1 holds.

A recap of today's lecture. Before I wrap up, what did we learn today? We learned that for real symmetric matrix A, it is positive definite, if for any given vector v, v^TAv is strictly positive for all non-trivial vectors v. And now, we also showed that this condition is equivalent to saying that all the eigenvalues of the matrix A are strictly positive. So, these are two equal ways of defining a positive definite matrix. Positive definite real symmetric matrix, we could show this equivalence rigorously.